

Variations on factivity in Hungarian

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Draft of March 21, 2025

Final version in the *Journal of Uralic Linguistics*, Volume 4, Issue 1, Jul 2025, p. 4-50,
<https://doi.org/10.1075/jul.00038.abr>

Abstract

In this paper I examine the empirical properties of so-called factive verbs in Hungarian, with an emphasis on various syntactic, morphological and pragmatic factors that induce factivity alternations with the same predicates. On the basis of these facts I argue that factivity is not a single property, but a set of properties: a speaker-related evidential veridical inference, a subject-related evidential veridical inference and the projection of the speaker-related veridical inference from the scope of entailment-cancelling operators. Although a default pragmatic principle favours the co-occurrence of these properties, morpho-syntactic and contextual cues can signal non-speaker-veridical or non-projective readings. The lexical meaning of so-called factives does not entail the truth of the complement, only suggesting that the attitude is based on a certain type of evidence.

Keywords factives, veridicality, attitude verbs, presuppositions, Hungarian

1 Introduction

Hungarian attitude verbs show various degrees of alternations in factivity conditioned by syntax, morphology, prosody and the context. The main empirical aim of this paper is to examine in detail this variation and understand under what conditions factive and non-factive interpretations can occur (cf. Kiefer 1978; Vendler 1980; Gyarmathy 2012; Ürögdi 2008; Cuba and Ürögdi 2010; Abrusán 2011 for previous discussion of various subsets of these facts).

As a result of this investigation, it will be shown that the property of factivity as usually understood is only a cover name for a set of properties that comprises two types of veridicality as well as projectivity. Veridicality is usually understood to mean that the truth of its complement follows from the truth of the entire sentence. This condition is met in (1a), but not in (1b):

- (1) a. *Peter knows that it is raining.* $\rightarrow$ *It is raining.*
 b. *Peter thinks that it is raining.* $\nrightarrow$ *It is raining.*

I show that in Hungarian the first example, (1a), has equivalents that can describe (at least) two different situations, one in which the truth of the complement follows for the attitude holder based on her evidence (I call this *subject-related evidential veridicality*) and one in which the truth of the complement follows from the evidence available for the speaker (and other conversational partners) (I call this *speaker-related evidential veridicality*). Note that the way I use the terms here, *speaker-related veridicality* is not simply belief that the complement is true, but belief based on reliable evidence, or justified belief. (cf. Özyildiz 2017a,b; Bondarenko 2023; Jeong 2020b for similar facts in Turkish, Azeri and Korean, see also Section 2.7. for discussion). The term *projection* is usually understood to mean that if a sentence is within the scope of an entailment-cancelling operator (e.g., negation), then the truth of the embedded clause typically still follows, in other words, the truth of the embedded clause projects. This is the case in (2a) but not in (2b):

- (2) a. *Peter doesn't know that it is raining.* $\rightarrow$ *It is raining.*
 b. *Peter doesn't think that it is raining.* $\nrightarrow$ *It is raining.*

I will note that projection is always the projection of the *speaker-related evidential veridical* inference, and not the projection of the *subject-related evidential veridical* inference. The properties of subject- and speaker-related veridicality and projectivity often co-occur, but not necessarily. Thus, the same verb might appear in morpho-syntactic contexts in which it is only subject-veridical, in some other cases in which it is subject and speaker veridical, but not projective, and yet in other cases where all conditions for “factivity” are met, speaker-veridicality and projection.

As a consequence of this observation, I argue that factivity is not a simple lexical property that some verbs have and others don't. It follows that in Hungarian there are no lexically factive and lexically non-factive verbs. Instead, the lexical property some attitude verbs have (including, but not limited to, the verbs usually called ‘factive’) is what I termed *subject-related empirical veridicality*: the property of the subject (attitude holder) having reliable evidence for the truth of the complement. The interpretation of verbs with this property is typically strengthened—if the syntax, morphology, prosody and context permit—to also imply speaker-related empirical veridicality. In many cases such strengthening is a default expectation, though a defeasible one. These default expectations may explain why some verbs have been termed ‘factive’ verbs. (Cf. also Roberts and Simons 2024 for a recent discussion of defaults in the context of factivity.)

Projection, I propose, is a complex pragmatic process that takes into account morpho-syntactic structure, event structure, information structural considerations, as well as the probability of conveyed information in context. I outline a proposal according to which projective inferences are hypotheses formulated by the listener about what facts the speaker takes as given. This hypothesis formation is probabilistic and is influenced by both grammatical and pragmatic factors. As a result, the inference that the embedded clause is true even within the scope of an entailment-cancelling operator is the outcome of an interplay of

multiple factors, such as context, information structure, event structure and the probability of the information being true.

The view defended in this paper contrasts with the traditional approach (cf. Kiparsky and Kiparsky 1970), according to which there are factive and non-factive verbs: attitude verbs belonging to the first group presuppose the truth of the embedded clause, while verbs in the second group do not. At least for Hungarian, this claim is too strong. Already Kiefer (1978), who in his groundbreaking work on Hungarian factive verbs argued in favour of factivity as a pre-encoded lexical property, noted that information structural factors also influence whether inferences typical of factivity hold. Kiefer resolved this duality by suggesting that even lexically non-factive verbs can give rise to pragmatically motivated factive inferences under certain conditions. Vendler (1980), despite assuming that the Hungarian verb *tud* ‘know’ is factive, made brief remarks about the verb *mond* ‘tell’, in which he noted that the veridical and projective inferences of this predicate are governed by morphosyntactic factors.¹

In this paper, I do not present a formal theory of how a probabilistic theory of projection can be implemented. An example of such a framework could be the RSA theory (cf. Qing et al. 2016; Warstadt 2022; Scontras and Tonhauser 2024 for work in RSA on presupposition.) Yet, a detailed examination of whether the ideas presented here can be captured in a framework such as RSA or other probabilistic pragmatic models will have to wait for another occasion. A final remark: This paper concentrates on the question of what factors condition the factive-alternations we observe, and how such variation might be explained. I will not address in detail what it means to be a ‘non-factive factive’, and how subject-related and speaker-related veridicality are connected. For the formal lexical semantics of *tud* ‘know’, see Abrusán (2024, 2025).

The paper is structured as follows: First, I turn to the description of the main data concerning the verb *tud* ‘know’. In Section 3, I examine many other verbs in Hungarian that can take propositional complements and discuss their veridical and projective inferences, if any. Section 4 looks more closely at the nature and position of the proleptic pronoun that we find in matrix clauses as well as the correlation of syntactic facts with ‘factivity’. Section 5 presents the main analysis proposed by this paper. Section 6 discusses examples of perspective shifted readings of factives and Section 7 concludes.

2 The verb *know*

Here is a seemingly simple question: Is the verb *tud* ‘know’ factive in Hungarian? Factivity, as usually understood, encompasses two properties: (a) *veridicality*: that the predicate entails the truth of its complement and (b) *projectivity*, that the truth of the complement is normally implied even in some entailment-cancelling contexts (e.g. under negation). The

¹For English, at least, experimental data also support the view that information structure (cf. Beaver 2010, Tonhauser 2016) and the probability of conveyed information (Degen and Tonhauser 2021) play an important role.

answer to the question, I will show below, is neither ‘yes’ nor ‘no’: factivity is a cover name for a set of properties that can occur independently of each other.

The property of veridicality itself is more complex than suggested by the traditional definition mentioned above: veridicality is understood as relative to an individual making the truth-assessment. As was stressed by De Marneffe et al. (2012), veridicality judgements are inferences made by *hearers* based on the lexical meaning of the predicate in question and certain assumptions about the speaker and the reported individuals; thus, veridicality judgements have a pragmatic component.² Veridicality, as defined in Giannakidou and Mari (2021) and previous work by the authors, refers to a judgement about the truth of a proposition made either by *the speaker* (what they call ‘objective veridicality’) or *the attitude holder* (what they term ‘subjective veridicality’). According to Giannakidou and Mari (2021), veridicality judgements can be made based on various grounds, that range from actual evidence to pure beliefs.

In the case of so-called factive verbs in Hungarian, veridicality judgements are made based on actual evidence, but can be anchored either only to the attitude holder or both to the attitude holder and the speaker. Accordingly, I define subject-related and speaker-related evidential veridicality:³

- (3) a. **Subject-related evidential veridicality:** The reported subject (attitude holder) has sufficient evidence / justification for the truth of the complement. Based on this, the subject takes the complement to be true.
- b. **Speaker-related evidential veridicality:** The speaker has sufficient evidence / justification for the truth of the complement. Based on this, the speaker takes the complement to be true.

The lexical meaning of many attitude predicates encodes the nature of the evidence that the attitude holder has: e.g. *see* encodes that the attitude holder has visual evidence, *realize* encodes evidence based on reasoning, etc. In the case of *tud* ‘know’ the type of the evidence or justification is unmarked, only the existence of some sort of evidence is required. The subject having a certain type of evidence (i.e. subject-related evidential veridicality) is always implied, as it is part of the lexical meaning of the verb. On top of this, speaker-related evidential veridicality can also be implied, but this is not always the case. At the same time, a double-sided speaker+subject-related evidential veridicality is the default understanding of many constructions—while purely subject-related evidential veridicality is often the marked case. When we find speaker+subject-related evidential veridicality, the hearer typically infers the truth of the complement as well. The reason

²Further, De Marneffe et al. (2012) argue that veridicality is not categorical but should be modelled as distributions.

³Giannakidou and Mari (2021) define ‘subjective veridicality’ as the attitude holder’s commitment to the truth of some proposition. Subject-related evidential veridical inference, as defined above, is not the same as subjective veridicality of Giannakidou and Mari (2021) as for them verbs such as *believe* are subjectively veridical. But the property defined in (3a) is not true of attitudes such as belief (whether understood as solipsistic or conjectural, in their terminology), that are not necessarily based on any actual evidence.

why speaker+subject-related evidential veridicality tends to be the default understanding of many so-called factive verbs is our tendency to think of others as sharing the same experience with the world as we do. Departure from this tends to be the marked case, at least when the existence of publicly available evidence is implied. NB: In this paper, the discussion of the lexical semantics of so-called factive verbs is purely descriptive. For a more detailed analysis of the lexical semantics of *tud* ‘know’, see Abrusán (2025).

Projection is an assumption made by the hearer about the state of the world based on what the speaker said:

- (4) **Projection** (of the complement of an attitude verb): the inference (drawn by the hearer) that the complement of the attitude verb is true in certain entailment-cancelling contexts.

In the following I discuss five cases that all involve the same verb *tud* ‘know’ but that give rise to different patterns of inferences. In the discussion I often abbreviate ‘subject-related evidential veridicality’ and ‘speaker+subject-related evidential veridicality’ as the properties of being subject-veridical and speaker+subject-veridical, respectively.

2.1 The textbook case

The first case is in tune with standard assumptions about factive verbs. The sentence in (5a) is speaker+subject-veridical and when the sentence is embedded under an entailment-cancelling operator (negation) the complement projects (5b):

- (5) a. *Anna tudja (azt), hogy Mari lemondott.*
 Anna know.3SG.DEF (it.ACC) that Mari resigned
 ‘Anna knows that Mari resigned.’
 b. *Anna nem tudja (azt), hogy Mari lemondott.*
 Anna not know.3SG.DEF (it.ACC) that Mari resigned
 ‘Anna doesn’t know that Mari resigned.’
 c. *...(De) Mari nem mondott le.*
 ...(but) Mari not resigned PRT
 ...‘(But) Mari did not resign.’
 # after (a) and # after (b)

Syntactically, the matrix sentence can optionally contain a pronominal element *azt* ‘it.ACC’ which forms a syntactic or referential chain with the embedded complement (cf. discussion in Section 4). This anaphor has the form of a definite pronoun, it is in the accusative case and triggers definiteness agreement on the verb. It is different from the complementizer *hogy*, which is obligatory and does not have a pronominal-looking form. Importantly, the pronoun *azt* ‘it.ACC’ in this case, if it appears, is postverbal. Moving it to a preverbal position is possible, but changes the nature of the example, as we will see below.

2.2 The anaphoric case

In the second example the postverbal propositional anaphor, *arról*, is in the sublative case, roughly meaning ‘about it’:

- (6) a. *Anna tud arról, hogy Mari lemondott.*
 Anna know.3SG.INDEF it.SUBL that Mari resigned
 ‘Anna knows about Mari’s resignation.’
 b. *Anna nem tud arról, hogy Mari lemondott (volna).*
 Anna not know.3SG.INDEF it.SUBL that Mari resigned (would)
 ‘Anna doesn’t know about Mari’s (supposed) resignation.’
 c. *...(De) Mari nem mondott le.*
 ...(but) Mari not resigned PRT
 ...‘(But) Mari did not resign.’
 # after (a), # after (b-indicative) and OK after (b-conditional)

The pronoun *arról* is obligatory and signals that the complement is discourse-old, i.e. its content has already been mentioned in the discourse. Since *arról* has an oblique case, it requires indefinite agreement on the verb *tud* ‘know’.⁴ The example in (6a) is speaker+subject-veridical. The example in (6b) can be in the indicative mood or in the conditional mood in the complement clause: with the indicative there is a projective inference that the complement is true (according to the speaker); with the conditional we find only the inference that the content of the complement has already been mentioned in the discourse. Although it has been claimed that this shows that (6b) is not factive [projective] (cf. Kiefer 1978), this conclusion does not follow. The meaning of the sentence in the conditional mood is a modal statement, namely that Mari’s resignation was possible. Presupposing this modal statement places only a very weak requirement on the context, in the above case, only that Mari’s resignation is possible. For this reason, (6b) is not a convincing example of non-projection.

2.3 The evidential case

Things start to get more involved when instead of the pronominal *azt* or *arról*, *tud* ‘knows’ stands with a preverbal adverbial pro-form *úgy* ‘so’, which is sometimes also classified as an evidential marker (cf. Kugler 2012). Examples with the pro-form *úgy* are neither speaker-veridical (7a), nor does the complement project in (7b), cf. also Kiefer (1978), Abrusán (2011), Gyarmathy (2012). NB: the adverbial pro-form *úgy* is obligatorily preverbal; it is ungrammatical in a postverbal position.

- (7) a. *Anna úgy tudja, hogy Mari lemondott.*
 Anna so know.3SG.DEF that Mari resigned

⁴On definiteness agreement with the object in Hungarian, see Bartos (2001), Bárány (2017), Coppock (2022), among others.

- ‘Anna (lit.) so-knows that Mari resigned.’⁵
- b. *Anna nem úgy tudja, hogy Mari lemondott (volna).*
 Anna not so know.3SG.DEF that Mari resigned (would)
 ‘Anna doesn’t (lit.) know so that Mari (would have) resigned.’
- c. *...(De) Mari nem mondott le.*
 ...(but) Mari not resigned PRT
 ...‘(But) Mari did not resign.’
 OK after (a) and OK after (b)

However, (7a) does have subject-related veridical inference: it reports that the subject experiences an attitude that they would describe as knowledge. (7b) on the other hand reports that the subject lacks such an attitude. Thus, the meaning of the sentences with *úgy tud* are not synonymous with simply reporting belief; they report the experience of knowledge on the part of the subject, and imply that the subject has sufficient evidence for their attitude, but the subject’s attitude is treated with a certain agnostic distance by the speaker.⁶ The example in (7a) is not anti-veridical: it is compatible with the truth of the complement, but does not require it. The example in (7b), out of the blue, has a weak anti-veridical implication that Mari did not resign, at least if Anna is a reliable source, but this implication is easily cancellable. In the scope of negation as in (7b) it is possible to find the conditional mood in the complement clause. In this case, there is a further inference that a question concerning the truth of the embedded complement has already been discussed.⁷ Finally, in the examples in (7) we obligatorily need to have definite object agreement on the verb *know*, which suggests the presence of a silent accusative-marked definite pronoun.

2.4 Preverbal evaluative adverbs

Another interesting case is modification of *know* with accented pre-verbal evaluative modifiers such as *jól* ‘well’ and *rosszul* ‘badly’, etc: (cf. also Gyarmathy 2012)⁸

- (8) a. *Anna JÓL tudja, hogy Mari lemondott.*
 Anna well know.3SG.DEF that Mari resigned
 ‘Anna knows correctly that Mari resigned.’
- b. *Anna ROSSZUL tudja, hogy Mari lemondott.*
 Anna badly know.3SG.DEF that Mari resigned
 ‘Anna (lit.) knows incorrectly that Mari resigned.’

⁵I resist translating the sentence with an English equivalent as I have not found one. In particular, candidate translations such as *believes to know* or *takes herself to know* have anti-veridical implications that the Hungarian sentence lacks.

⁶(7) expresses a stronger commitment, as far as I can see, than examples of so called ‘conjunctural belief’ discussed by Giannakidou and Mari (2021). A detailed comparison, unfortunately, has to be postponed.

⁷Without negation, the conditional mood is not possible in the antecedent of a conditional, though it is marginally available in questions.

⁸I assume that in these cases the evaluative adverb is focused, which I mark with capitals, for the sake of readability.

- c. *Anna nem JÓL tudja, hogy Mari lemondott.*
 Anna not well know.3SG.DEF that Mari resigned
 ‘Anna doesn’t (lit.) know correctly that Mari resigned.’
- d. *...(De) Mari nem mondott le.*
 ...(but) Mari not resigned PRT
 ...(But) Mari did not resign.
 # after (a), OK after (b), OK after (c)

The example in (8a) is speaker+subject-veridical, but in a complex way: it conveys subject-related veridicality and also that the speaker agrees with the subject. The example in (8b) is anti-veridical in the following sense: it conveys a subject-related veridical inference but also that the speaker thinks that the subject is wrong. Example (8c) has the same anti-veridical implication as (8b); thus the veridical inference of (8a) does not project.

In the above examples, it is important that the evaluative adverb is preverbal and accented. The same adverbs in postverbal position, if they are acceptable at all, only have an intensifier or discourse particle meaning, similar to *know well* in English:⁹

- (9) a. *Anna tudja jól, hogy Mari lemondott (mégis úgy tesz, mintha nem tudná).*
 Anna know.3SG.DEF well that Mari resigned (still so does, as.if not knew)
 ‘Anna knows well that Mari resigned (she still behaves as if she didn’t know).’
- b. **Anna tudja rosszul, hogy Mari lemondott.*
 Anna know.3SG.DEF badly that Mari resigned
 ‘Anna knows badly that Mari resigned.’
- c. **Nem igaz, hogy Anna tudja jól, hogy Mari lemondott.*
 not true that Anna know.3SG.DEF well that Mari resigned
 ‘It is not true that Anna knows well that Mari resigned.’

The example (9a) above is speaker+subject-veridical in a similar sense as (5a), it implies a shared knowledge of truth between subject and speaker. However, it is not possible to negate it, as shown in (9c).

2.5 Focus on the pronoun *azt* ‘it.acc’

The next case I discuss is a variant of (5), but involves focus on the proleptic pronoun *azt* ‘it.ACC’:

- (10) a. *Anna AZT tudja, hogy Mari lemondott.*
 Anna it.ACC know.3SG.DEF that Mari resigned
 ‘What Anna knows is that Mari resigned.’

⁹The intensifier/discourse particle meaning is also available with non-accented preverbal *jól* and in this case *jól* can be modified with *nagyon* ‘very’.

- b. *Anna nem AZT tudja, hogy Mari lemondott.*
 Anna not it.ACC know.3SG.DEF that Mari resigned
 ‘What Anna knows is not that Mari resigned.’

Focusing in Hungarian involves syntactic movement to an immediately preverbal position, cf. e.g. Szabolcsi (1981), É. Kiss (2002), Szendrői (2001). In the examples above, the pronoun *azt* ‘it.ACC’ immediately precedes the verb *tud* ‘know’ and bears focus accent. Although (10a) is speaker+subject-veridical, (10b) does not require a projective inference, though it is compatible with it. The actual interpretation of the sentences depends heavily on the context. Imagine first the following context:

- (11) *Context 1:* The information state includes that János was fired and that Mari resigned. The relevant question is: what does Anna know from all this?

In this context, the truth of the complement is already entailed by the context; the positive sentence in (10a) is compatible with this, and the negative sentence in (10b) is also compatible with this. The facts are similar (in fact, clearer) with a syntactically unambiguous wide-scope negation:

- (12) *Nem igaz, hogy Anna AZT tudja, hogy Mari lemondott.*
 not true that Anna it.ACC know.3SG.DEF that Mari resigned.
 ‘It is not true that what Anna knows is that Mari resigned.’

Thus, in this context, the interpretation of the positive sentence can be said to be speaker+subject-veridical. The acceptability of the negative form is compatible with the complement being projective.

Observe now the following context:

- (13) *Context 2:* The information status does not contain any other information, only the question *What does Anna know?*

The sentence in (10a) implies the truth of the complement, though this information is not entailed by the context. The truth of the complement of the verb in the negative sentence in (10b), on the other hand, is not necessarily understood to project; it is compatible with the follow-up *because she did not resign*. This being said, a felicitous use of the sentence does normally require that the question of Mari resigning has been at least mentioned or treated as a possibility in the context, otherwise talking about Anna’s attitude to it would not be relevant. The important point is that there is no obligatory inference to the truth of the complement in (10b). (Again, the facts are even clearer with unambiguously wide-scope negation as in (12).¹⁰)

The examples with preverbal focus on the pronominal anaphor can sometimes also be found in cases with a perspective shifted reading, and indeed contrastive stress seems to

¹⁰ A reviewer disagrees with me on the empirical point that Context 2 does not require projection. The consultants I have asked can get a non-projective reading, but proper empirical tests would be needed to firmly establish this claim.

facilitate this reading. I come back to these cases in Section 6.

2.6 *Know* in Hungarian: Summary

Table 1 summarises the local environments in which the verb *know* can appear:

1.	Anna	tudja	(azt) ‘it.ACC’	hogy ...
2.	Anna	tud	arról ‘about it’	hogy...
3.	Anna	úgy ‘so’	tudja	hogy...
4.	Anna	JÓL ‘well’	tudja	hogy...
5.	Anna	AZT ‘it.ACC’	tudja	hogy...
Anna		knows		that...

Table 1: Summary of the local environments for the verb *know*

As was reviewed above, the first two cases, in which the pronominal anaphor is postverbal, are speaker+subject-veridical. The third and fourth cases, in which there is an adverbial pro-form or a modifier in the (immediately) preverbal position are not necessarily speaker+subject-related veridical or projective, though a subject-related veridical inference always persists. The fifth case is speaker+subject-veridical, but projection can be influenced by focus. There is thus a clear pattern: non-speaker-veridical or non-projective readings are associated with an emphasis on an element in the (immediately) preverbal field that relates to the complement.

The traditional property of *factivity* is in fact a combination of (at least) three properties (a) a subject-related evidential veridical inference, (b) a speaker-related evidential veridical inference and (c) projection of the complement clause.¹¹ In so-called factive verbs, and in particular in the case of *tud*, we typically find both speaker and subject-related veridical inferences at the same time, but in some cases, as with *úgy*, only a subject-related evidential veridical inference can be observed.

2.7 Comparisons: Korean, Turkish, Azeri

Syntactically, morphologically or prosodically conditioned factivity alternations have been observed in many languages. One potential factor is the presence or absence of the subjunctive in the complement clause (as in Catalan, Greek, cf. Giannakidou (1998) and also Abrusán (2011) for discussion). Another type of data was observed more recently in Turkic languages (Turkish, Azeri, cf. Özyildiz 2017a,b; Bondarenko 2023) and Korean (Jeong 2020b). I turn to these here.¹²

¹¹It is also sometimes assumed that the veridical/projective inference, like other presuppositions, should be already given in the context. Nevertheless, this is a contentious matter in the case of factives, cf. e.g. Spenader (2002).

¹²Thanks to an anonymous reviewer for pressing me to address these similarities.

Turkish Özyildiz (2017a,b) discussed attitude verbs such as *bil-* ‘know’ in Turkish. These verbs can take either a nominalized complement or a tensed clause; veridical readings are observed only with nominalized complements, as long as the nuclear pitch falls on the matrix predicate. If, however, the nuclear pitch accent falls on a constituent of the embedded nominalized clause, the resulting string-identical sentence can be interpreted as non-factive.

In contrast, all the examples with so-called factive verbs in Hungarian discussed above involved tensed complement clauses, whether or not they received a factive interpretation. Although nominalized complements to attitude verbs do exist in Hungarian, similarly to English, they have a rather restricted distribution and depend on the availability of a possessive nominal construction. Nominalized complements with *tud* are interpreted as veridical (regardless of focus); however, with other verbs (e.g. *mond* ‘say’) nominalized complements can be interpreted in a non-veridical fashion.

Korean Jeong (2020a,b) showed that the veridicality of the verbs *know*, *remember* in Korean systematically depends on prosody even in unembedded contexts. The pattern is similar to Turkish, but the sentences showing prosodically conditioned alternation in veridicality do not involve nominalization.

The focus sensitivity of Hungarian *tud* ‘know’ shows a different pattern. In the case of *tud* ‘know’ focus on a constituent of the embedded clause (or focus on the propositional anaphor *azt*) does not eliminate veridicality, though there might be an effect on projection, as mentioned above. However, as will be discussed in Section 3.1, verbs such as *emlékszik* ‘remember’, perception verbs such as *lát* ‘see’, *hall* ‘hear’ and verbs such as *sejt* ‘suspect’ do show a variation in their veridicality depending on whether the pronominal anaphor is in a preverbal accented position or in a postverbal de-accented position.

Azeri Bondarenko (2023) discusses factivity-alternations in Azeri, a Turkic language. She shows that Azeri has the same type of preverbal nominalized clauses as Turkish, which behave in similar ways to Turkish. But she shows that Azeri also has another type of complement clause that can appear with attitude verbs, not discussed by Özyildiz (2017a,b), (non-nominalized) clauses formed with the complementizer *ki*. This type of complement can co-occur with a proleptic pronoun *elə* that seems to be very similar to *úgy* in Hungarian. In this case, the readings obtained are non-veridical. *Ki*-clauses can also occur without this anaphor (or with a case-marked pronominal *o*, which, at least superficially, resembles *az* in Hungarian).

Thus, Turkish, Korean and Azeri show many similarities but also differences with the data in Hungarian. It remains to be seen whether the type of analysis proposed in this paper can be applied to these languages as well.

3 Some other factives and non-factives in Hungarian

In this section I review some other verbs that alternate in their veridicality and projection. (cf. also Tóth 2006 and Gyarmathy 2012 for a discussion of the range of verbs that can appear with *úgy*).

3.1 Verbs compatible with *úgy*

Most attitude verbs in Hungarian are compatible with the adverbial pro-form *úgy*, except emotive factives and verbs with a verbal particle (discussed in the next subsection). In particular, *úgy* can also modify some other verbs that can have factive readings, principally the Hungarian equivalents of cognitive verbs (e.g. *remember*, *understand*), perception verbs (*hear*, *see*) but also the verbs *inform* and *notice*:

- (14) *úgy* is compatible with
emlékszik ‘remember.3SG’, *érti* ‘understand.3SG’, *látja* ‘see.3SG’, *hallja* ‘hear.3SG’,
tájékoztat ‘inform.3SG’, *veszi észre* ‘notice.3SG’

When used with *úgy*, the above verbs are not speaker-veridical nor projective, which I illustrate with the verb *lát* ‘see’:

- (15) a. *Jancsi úgy látta, hogy esik az eső, de tévedett.*
 Jancsi so saw.3SG that falls the rain but was.mistaken
 ‘Jancsi had the visual perception (lit. so-saw) that it was raining, but he was mistaken.’
 b. *Jancsi nem úgy látta, hogy esik az eső, és nem is esett.*
 Jancsi not so saw.3SG that falls the rain and not either rained
 ‘Jancsi didn’t have the visual perception (lit. didn’t so-see) that it was raining, and indeed it didn’t rain.’

These verbs can also be modified with *rosszul* ‘badly/incorrectly’:

- (16) *Jancsi rosszul látta hogy esik az eső.*
 Jancsi badly saw.3SG that falls the rain
 ‘Jancsi had the incorrect visual perception (lit. incorrectly so-saw) that it was raining’

The same verbs, when they stand with a postverbal pronominal anaphor (which might be omitted), are speaker+subject-veridical and projective:

- (17) *emlékszik arra* ‘remember.3SG it.SUBL’, *érti azt* ‘understand.3SG it.ACC’, *látja azt* ‘see.3SG it.ACC’, *hallja azt* ‘hear.3SG it.ACC’, *tájékoztat arról* ‘inform.3SG about it’), *észreveszi azt* ‘notice it.ACC’
 (18) a. *Jancsi látta (azt) hogy esik az eső, #de tévedett.*
 Jancsi saw.3SG it.ACC that falls the rain but was.wrong
 ‘Jancsi saw that it was raining, but he was wrong.’
 b. *Jancsi nem látta (azt) hogy esik az eső, #és nem is esett.*
 Jancsi not saw.3SG it.ACC that falls the rain and not either rained
 ‘Jancsi saw that it was raining, and indeed it didn’t rain.’

Moving the pronominal anaphor to a pre-verbal position, however, makes a non-factive

reading possible:¹³

- (19) *Jancsi azt látta hogy esik az eső, de tévedett.*
 Jancsi it.ACC saw.3SG that falls the rain but was.wrong
 ‘Jancsi saw that it was raining, but he was wrong.’

We can add to the above that in the case of certain other cognitive or perception verbs that are not usually considered factive, emphasizing the verb (i.e. putting the pronominal anaphor in a postverbal position) is associated with factive inference. This is observed, for instance, with the verb *sejt* ‘suspect’ (illustrated in (20a)) and the verb *érez* ‘feel’. However, if the preverbal modifier position is occupied by the propositional anaphor *azt* or adverbial modifiers, the interpretation is non-factive, in the sense that the truth of the embedded clause is not speaker-veridical in unembedded cases and does not project in the scope of entailment-cancelling operators (cf. (20b-c) for *sejt* ‘suspect’):

- (20) a. *Jancsi sejti (azt), hogy Mari megcsalja (#de Mari valójában hűséges).*
 Jancsi suspects (it.ACC) that Mari cheats.on.him (but Mari in.reality faithful)
 ‘Jancsi suspects that Mari is cheating on him (#but in fact she is faithful).’
 b. *Jancsi azt/úgy sejti, hogy Mari megcsalja (de Mari valójában hűséges).*
 Jancsi it.ACC/so suspects that Mari cheats.on.him (but Mari in.reality faithful)
 ‘Jancsi suspects that Mari is cheating on him (but in fact she is faithful).’
 c. *Jancsi rosszul sejti, hogy Mari megcsalja (mert Mari valójában hűséges).*
 Jancsi wrongly suspects that Mari cheats.on.him (because Mari in.reality faithful)
 ‘Jancsi is mistaken in suspecting that Mari is cheating on him (because in fact she is faithful).’

Table 2 summarises the syntactic environments of the verbs *lát* ‘see’ and *sejt* ‘suspect’. Emphasis on the verb (i.e. postverbal pronominal anaphor) results in factive readings. The presence of the adverbial pro-form *úgy* and adverbial modification with *rosszul* ‘badly’ is

¹³A naturally occurring example:

- (i) *Késelni kezdett egy vonaton, miután azt látta, hogy egy utas démonná változott.*
 knife.INF started a train.SUE after it.ACC saw.3SG that a passenger demon.TRANS mutated
 ‘He started to knife on a train after he saw a passenger turn into a demon.’
 (<https://parameter.sk/339193/keselni-kezdet-egy-vonaton-miutan-azt-latta-hogy-egy-utas-demonna-valtozott> (accessed on October 2, 2024))

associated with non-speaker-veridical readings. Moving the pronoun *azt* to a pre-verbal position is compatible with non-veridical readings (though it does not require it).

1.	Péter		látja/sejti	(azt) ‘it.ACC’	hogy ...
2.	Péter	úgy ‘so’	látja/sejti		hogy...
3.	Péter	rosszul ‘badly’	látja/sejti		hogy...
4.	Péter	azt ‘it.ACC’	látja/sejti		hogy...
	Péter		sees/suspects		that...

Table 2: Summary of the local environments for the verb *sejt* ‘suspect’

Notice finally that doxastic verbs such as *gondol* ‘think’ and *hisz* ‘believe’, as well as the verbs *képz* ‘imagine’, *tart* ‘assume’ typically appear with *úgy* or with a preverbal pronominal anaphor but not with postverbal pronominal anaphors, cf. Ürögdi (2008, 2012) and Cuba and Ürögdi (2010), discussed in Section 4.2:

- (21) *Jancsi azt/úgy gondolta hogy esik az eső, de tévedett.*
 Jancsi it.ACC/so thought.3SG that falls the rain but was.wrong
 ‘Jancsi thought that it was raining, but he was wrong.’

With these verbs the pronominal anaphor can only appear postverbally if the verb itself is focused (cf. example (22)) or if another constituent appears in a preverbal position as in (23). Yet, this does not result in a veridical reading:

- (22) *Nadal minden nap [gondolt]_F arra, hogy itt a vége.*
 Nadal every day thought it.SUBL, that here the end.POSS
 ‘Nadal had the thought every day that this was the end.’¹⁴
- (23) *[Egy egész generáció]_F hiszi azt, hogy dolgozni időpocsékolás.*
 a whole generation believes it.ACC that work.INF waste.of.time
 ‘A whole generation believes that working is a waste of time.’¹⁵

3.2 Veridical verbs not compatible with *úgy*

Perfective veridicals The adverbial pro-form *úgy* normally cannot appear with typically speaker+subject-veridical verbs that have a perfective particle (which I mark with a dot: PRT.verb).¹⁶

- (24) **úgy* with

¹⁴<https://www.eurosport.hu/tenisz/australian-open/2022/nadal-minden-nap-gondolt-arra-hogy-itt-a-vege-vid1611379/video.shtml> (accessed on October 2, 2024)

¹⁵<https://www.facebook.com/watch/?v=1389130268334275> (accessed on October 2, 2024)

¹⁶Gyarmathy (2012) mentions two exceptions: *észre.vesz* ‘notice’ and *meg.állapít* ‘establish’, however, for me, the second is not acceptable with *úgy* nor have I found any examples on the web, while the first is not a canonical verbal particle.

rá.jön ‘realize’, *rá.döbben* ‘realize with a shock’, *fel.fedez* ‘discover’, *el.felejt* ‘forget’, *be.vall* ‘admit’, *fel.tár* ‘disclose’, *meg.állapít* ‘establish’, *meg.erősít* (‘confirm’)

The verbs in (24) typically have speaker+subject-veridical readings. (Many of the above verbs belong to a group of verbs that are sometimes called semi-factives—cf. e.g. Karttunen 1971, Stalnaker 1974, Chierchia and McConnell-Ginet 2000—though the existence of such a category has been disputed.) The projection of the speaker-veridical inference, as was first noted by Kiefer (1978), in Hungarian is heavily influenced by the focus structure of the sentence. For example, according to Kiefer, the following two sentences have different interpretations:¹⁷

- (25) a. [*Arra*]_F *jött rá Péter, hogy hibát követett el?*
 it.SUBL came PRT Péter that mistake.ACC committed PRT
 ‘Did Péter realize that he made a mistake?’
 b. [*Rájött*]_F *Péter arra, hogy hibát követett el?*
 PRT.came Péter it.SUBL that mistake.ACC committed PRT
 ‘Did Péter realize that he made a mistake?’

In (25a), the main emphasis—Kiefer argues—falls on the subordinate clause rather than the main clause verb; this is shown by the proleptic pronoun *arra* being in focus position. This example doesn’t necessarily imply the truth of the complement of the matrix verb. In (25b), however, the main emphasis falls on the verb of the main clause, which is focused, typically implying the truth of the complement.

According to Kiefer (1978), the factive presupposition appears in these cases when the subordinate clause is the topic of the entire sentence or at least a part of it. His conclusion is that there is no semi-factive category, but rather the factive alternation is a consequence of changes in information structure: if the emphasis is on the verb, a factive interpretation is obtained, whereas if it is on the proleptic pronoun (or the whole subordinate clause), it is not.

Curiously, perfective particles seem to be incompatible with *úgy* even with verbs that otherwise can appear with this pro-form. Thus, the perfective versions of the verbs in (14) are not compatible with *úgy*:

- (26) **úgy* with
meg.emlékezik ‘recall/mention.3SG’, *meg.ért* ‘come to understand.3SG’, *meg.lát*
 ‘get to see.3SG’, *meg.hall* ‘get to hear.3SG’, *meg.érez* ‘get to feel’, *rá.érez* ‘sense.3SG’

Further, the verbs *think* and *know*, when occurring with the particles *meg-/el-*, are not compatible with the adverbial pro-form *úgy*, though they are compatible with focused proleptic pronouns. (The particle is postverbal if other elements occupy the immediately preverbal position, as will be discussed in the next section.)

¹⁷Some of Kiefer’s (1978) examples involve focus inside the subordinate clause.

According to my intuition, the whole subordinate clause needs to be focused in these cases to get the non-factive interpretation in example (a).

- (27) a. *Mari *úgy/^{ok}[azt]_F tudta meg, hogy esik.*
 Mari so/it.ACC knew PRT that rains
 ‘That it was raining was what Mari got to know.’
 b. *Jancsi *úgy/^{ok}[azt]_F gondolta meg, hogy mit mondjon.*
 Jancsi so/it.ACC thought PRT that what say.SUBJ.3SG
 ‘What to say was what Jancsi contemplated about.’

A correlation between veridicality/factivity and perfectivity was also noted by Zuchewicz (2020) in Polish for many of the same verbs discussed in this subsection. In the following subsection I will also point out another interaction between veridicality and the presence of a perfective particle with communication verbs, and tentatively suggest an explanation. There is, however, much that remains to be understood about the interaction of veridicality and aspectual structure.

Emotives Emotives are not compatible with preverbal *úgy*, except when it is used as an intensifier (e.g. *úgy sajnálja* ‘regrets so much’), this use, however, does not interfere with veridicality. Kiefer (1978) observed that there are two types of emotive factives in Hungarian, exemplified with the two versions of the verbs *regret* below:

- (28) a. *sajnálkozik (azon)* ‘regret.REFL.3SG it.SUE’
 b. *sajnálja (azt)* ‘regret.3SG it.ACC’

The first type of emotives, which are almost all morphologically marked as reflexive, are not necessarily speaker-veridical, especially if the propositional anaphor is null or preverbal. With a postverbal proleptic pronoun usually a speaker-veridical reading is attested (as was noted by Kiefer).

- (29) a. *Mari (azon) sajnálkozik, hogy Péter mérges rá (pedig nem is*
 Mari (it.SUE) regret.REFL.3SG that Péter angry at.her but not either
mérges).
 angry
 ‘Mari is lamenting that Péter is angry at her, (though he is not).’
 b. *Mari sajnálkozik azon, hogy Péter mérges rá (#pedig nem is*
 Mari regret.REFL.3SG it.SUE that Péter angry at.her but not either
mérges).
 angry
 ‘Mari is lamenting/regrets that Péter is angry at her, (#though he is not).’

The contrast between (29a) and (29b) above led Kiefer (1978) to suggest that emotives of the *sajnálkozik* type are not factive, their apparent factivity is only a discourse effect of topicality.

The second kind of emotives, exemplified by *sajnál*, are more robustly veridical and projective, whether or not the accusative-marked proleptic pronoun is spelled out overtly.

- (30) *Mari sajnálja (azt), hogy Péter mérges rá (#pedig nem is mérges).*
 Mari regret.3SG it.ACC that Péter angry at.her but not either angry
 ‘Mari regrets that Péter is angry at her, (#though he is not).’

As for projection, Abrusán (2011) claimed that focusing the proleptic pronoun might interfere with projection in these cases as well, similarly to the class of verbs sometimes called semi-factives discussed above.¹⁸ Sentence (31a), in which the pronoun *azt* is in focus, presents a coherent statement, indicating that factive inference in this case either does not project or was not formed in the first place. However, sentence (31b), where the verb itself is focused, suggests a contradiction: here, the truth of the embedded clause does project.

- (31) a. *Ha Péter [azt]_F sajnálja, hogy rosszul vizsgázott, akkor*
 if Péter it.ACC regrets.3SG that badly took.the.exam.3SG then
feleslegesen bánkódik, mert jó jegyet kapott.
 unnecessarily is.sad because good grade.ACC got.3SG
 ‘If what Péter regrets is that he performed poorly on the exam, then he is sad unnecessarily because he received a good grade.’
 b. *#Ha Péter [sajnálja]_F, hogy rosszul vizsgázott, akkor feleslegesen*
 if Péter regrets.3SG that badly took.the.exam.3SG then unnecessarily
bánkódik, mert jó jegyet kapott.
 is.sad because good grade.ACC got.3SG
 #‘If Péter [regrets]_F that he performed poorly on the exam, then he is sad unnecessarily because he received a good grade.’

Let me note that focus seems to play a role in the interpretation of factive verbs in other languages as well, cf. Beaver (2010), Tonhauser (2016) (pace Djärv and Bacovcin 2020) for English, Özyildiz (2017a) for Turkish, Bondarenko (2023) for Azeri.

3.3 Verbs of communication

In English it is usually assumed that communicative verbs are not factive (cf. Anand and Hacquard 2014), though they may behave like factives in certain contexts (cf. Schlenker 2010). In Hungarian, however, as was observed by Vendler (1980) and Kiefer (1986), the perfective version of communicative verbs is factive:

- (32) *Kati azt mondta Péternek, hogy el akar válni. De nem akar*
 Kati it.ACC told Péter.DAT that PRT wants divorce.INF but not want.3SG
elválni.
 divorce.INF
 ‘Kati told Péter that she wants to divorce him. But she does not want to divorce.’

¹⁸According to a reviewer, there seems to be some variation in intuitions concerning these data.

- (33) *Kati mondtá Péternek, hogy el akar válni. ?De nem akar*
 Kati told Péter.DAT that PRT wants divorce.INF but not want.3SG
elválni.
 divorce.INF
 ‘Kati told Péter that she wants to divorce him. But she does not want to divorce.’
- (34) *Kati el.mondta Péternek, hogy el akar válni. #De nem akar*
 Kati PRT.told Péter.DAT that PRT wants divorce.INF but not want.3SG
elválni.
 divorce.INF
 ‘Kati told Péter that she wants to divorce him. But she does not want to divorce.’

Thus while the positive version of the example above in (34a) is normally not speaker-veridical, (34b) can be veridical depending on context and (34c) has a strong speaker-veridical inference. Moreover, the negative version of (34c) is projective as well:

- (35) *Nem igaz hogy Kati el.mondta Péternek, hogy el akar válni. #Nem*
 not true that Kati PRT.told Péter.DAT that PRT wants divorce.INF not
akar elválni.
 want.3SG divorce.INF
 ‘It is not true that Kati told Péter that she wants to divorce him. #She does not want to divorce.’

We find similar facts with other verbs of communication (36a), and the process seems to be productive with the particle *el-*, as shown in (36b):

- (36) a. *el.mondta* ‘told’, *meg.írta* ‘wrote’, *meg.jósolta* ‘predicted’, *el.árulta* ‘disclosed’,
le.közölte ‘published’
 b. *el.énekelte* ‘sang’, *el.suttogta* ‘whispered’, *el.krákogta* ‘croak’, *el.kürtölte* ‘tell everyone’, etc.

Why is the perfective particle connected to factivity? I tentatively suggest that the explanation can be found in the relation of communicative verbs to *definiteness effect* (DE) verbs (cf. Szabolcsi 1986, Bende-Farkas 1995, É. Kiss 1995, Kálmán 1995, Peredy 2009,...).

In Hungarian, verbal particles can have various semantic roles. According to É. Kiss (2008b), the primary role of so-called perfective verbal particles is to indicate a complex event structure. Let’s compare the following two sentences:

- (37) a. *Zsuzsa szaladt a szobába.*
 Zsuzsa ran the room.into
 ‘Zsuzsa was running into the room.’
 b. *Zsuzsa be.szaladt a szobába.*
 Zsuzsa PRT.ran the room.into
 ‘Zsuzsa ran into the room.’

The first sentence describes a process, while the second sentence describes a completed event, resulting in the state of being in the room. According to É. Kiss (2008b), while the verb on its own describes the process, the verb with the particle describes the complex event composed of a process and a result.

The situation is somewhat different for verbs whose object comes into existence as a result of the action. In the example below, both sentences denote a completed event:

- (38) a. *Péter írt egy könyvet.*
Péter wrote a book.ACC
‘Péter wrote a book.’
b. *Péter meg.írt egy könyvet.*
Péter PRT.wrote a book.ACC
‘Péter wrote a book.’

While in (38a) the main point concerns the creation of the result, in (38b) the main point is that the process has concluded, implying the existence of the result, at least in an abstract form (cf. É. Kiss 2008b, Piñón 2008a,b).

These verbs are also called Definiteness Effect (DE-) verbs. Canonical DE-verbs imply that their theme comes into existence as a result of the action described (cf. (38a)). The combination of a DE-verb with a perfective particle implies change of state or place affecting a specific theme, whose existence (in some form) is presupposed (cf. (38b), É. Kiss 2008b).

How is it possible to presuppose the existence of an object just about to be created? This contradiction is usually resolved by assuming that only the existence of an abstract object (or discourse oldness) is presupposed (cf. Bende-Farkas 1995, É. Kiss 1995, Kálmán 1995, Peredy 2009).

Observe now that many communicative verbs are at the same time DE-verbs:

- (39) a. *énekelt egy dalt* ‘sang a song’, *írt egy verset* ‘wrote a poem’, *mondott egy viccet* ‘told a joke’
b. *el.énekelt egy dalt* ‘sang a song’, *meg.írt egy verset* ‘wrote a poem’, *el.mondott egy viccet* ‘told a joke’

Based on this observation, I suggest that the factivity of the embedded clause is tied to the anaphoric nature of the perfective particle+DE verb combination. If the particle signals that the complement is discourse old, it is natural for the speaker to assume that it is also veridical. Moreover, the preverbal position of the particle also biases the hearer to assume that the complement is backgrounded, as the question of the completion of the event is in the foreground. The combination of the two effects creates a strong expectation for speaker-veridicality and projection.¹⁹

¹⁹Thanks to Edgar Onea for pressing me to clarify this point.

4 Propositional anaphors and information structure

4.1 The proleptic pronoun

In Hungarian, the main proposition of a sentence is encoded in the verb and the constituent immediately preceding the verb. According to the analysis of É. Kiss (2002, 2008b), the extended syntactic domain of the verb is the PredP, whose specifier position can be occupied by the focus constituent, the verbal particle, and other verb modifiers, as illustrated by the examples in (40) below. The main prosodic accent of the sentence falls on the left edge of the extended domain of the verb (see Szendrői 2001); that is, on the constituent in the specifier of PredP, or in its absence, on the verb itself. In other words, the domain of the specifier and head of PredP constitutes the informational focus of the sentence.

- (40) a. [TopP *János* [PredP *ki* [Pred' *nyitotta*_i [VP *t_i az ajtót*]]]]
 [TopP *János* [PredP *out* [Pred' *opened*_i [VP *t_i the door*]]]]
 'János opened the door.'
- b. [TopP *János* [PredP *kezet* [Pred' *fogott*_i [VP *t_i Zsuzsával*]]]]
 [TopP *János* [PredP *hand* [Pred' *shook*_i [VP *t_i Zsuzsa.with*]]]]
 'János shook hands with Zsuzsa.'
- c. [TopP *János* [PredP *PÉTERNEK* [Pred' *írt*_i [VP *t_i levelet*]]]]
 [TopP *János* [PredP *PÉTER.DAT* [Pred' *wrote*_i [VP *t_i letter.ACC*]]]]
 'János wrote a letter to Péter.'

According to Kenesei (1994), the pronoun *azt* 'it.ACC' (and also *arra* 'it.SUBL', *azon* 'it.SUE', etc.) is an expletive that forms a chain with the embedded clause. This account has been challenged by Tóth (2000), who argued that this element is a referential pronoun, functioning as a propositional anaphor (cf. also Rákosi et al. 2005, Szűcs 2015) and Den Dikken (2017) who proposed that *azt* should be viewed as a secondary predicate for the complement clause. In this paper I follow Tóth (2000) in assuming that the proleptic demonstrative *azt* is a referential pronoun, interpreted as a propositional anaphor, with the CP generated in an adjunction structure.²⁰ Syntactically, this pronoun can occupy various positions: it can be postverbal as in (41a), it can be in a preverbal modifier position (41b) or in other preverbal positions, e.g. focus (41c) or even contrastive topic (41d):

- (41) Glosses: *gondolta* 'think.3SG.PAST.DEF'; *azt* 'it.ACC'; *hogy* 'that'; *esik* 'rains'
- a. *Anna* [PredP [Pred' *gondolta*_j [VP *t_j (azt_i/t_i)*]]], [CP *hogy esik*]_i
 'Anna thought that it was raining.'
- b. *Anna* [PredP [Pred' *azt_i gondolta*_j [VP *t_j t_i*]]], [CP *hogy esik*]_i
 'Anna thought that it was raining.'
- c. *Anna* [PredP *AZT_i* [Pred' *gondolta*_j [VP *t_j t_i*]], [CP *hogy esik*]_i
 'Anna thought that it was raining (and not snowing, e.g.).'
- d. [_cTP *Azt_i* [PredP *Anna* [Pred' *gondolta*_j [VP *t_j t_i*]], [CP *hogy esik*]]_i

²⁰Thanks for the editors for helpful comments on this issue

‘That it was raining, it was Anna who thought (that).’

Sentences in which the proleptic pronoun is postverbal and is in the accusative case (*azt*) are synonymous with sentences with no proleptic pronoun, i.e. postverbal accusative propositional anaphors are optional. (Following É. Kiss 2002, I assume that PredP is an extended VP projection hosting verbal modifiers and focus.) The various syntactic positions for the proleptic pronoun are correlated with interpretive differences. I return to some of these differences below.

4.2 The position of the proleptic pronoun

Ürögdi (2008, 2012) and Cuba and Ürögdi 2010 (see also Molnár 2015) observed that (verbs traditionally assumed to be) factive attitude verbs are incompatible with preverbal proleptic pronouns, unless the latter are in focus (or other higher positions). In other words, the pronoun *azt* (‘it.ACC’) can only appear in the pre-verbal modifier position with non-factive verbs (cf. (41) above), but not with factives:

(42) Glosses: *tudja* ‘know.3SG.past.def’; *azt* ‘it.ACC’; *hogy* ‘that’; *esik* ‘rains’

- a. **Péter* [_{PredP} [_{Pred} [*azt*]_i *tudja*_j [_{VP} *t_j* *t_i*]]], [_{CP} *hogy esik*]_i
‘Péter knows that it is raining.’
- b. *Péter* [_{PredP} *AZT*_i [_{Pred} *tudja*_j [_{VP} *t_j* *t_i*]]], [_{CP} *hogy esik*]_i
‘Péter knows that it is raining (and not snowing, e.g.).’

Given this observation, the above authors suggest that the placement of informational focus in complex sentences (i.e., what the main assertion is) is related to factivity. According to their proposal, in the case of factive verbs, the matrix verb corresponds to the main assertion of the sentence, and the subordinate clause is merely mentioned. However, in the case of non-factives, the subordinate clause bears the main assertion of the sentence.

According to Cuba and Ürögdi’s (2010) syntactic analysis, there are two types of complements: referring ([CP]) and non-referring ([cP [CP]]) complements. The pronoun *azt* originates in the cP position of the complement and moves from there to the preverbal position of the matrix verb. For this reason, *azt* in the verbal modifier position can only appear with predicates that select for non-referring ([cP [CP]]) complements, but not with predicates that select for referring ([CP]) complements. So-called factive verbs select referring ([CP]) complements, and therefore they cannot co-occur with *azt* in a preverbal modifier position. Non-factive verbs, on the other hand, can select for either referring ([CP]) or non-referring ([cP [CP]]) complements, making the presence of *azt* optional. The absence of *azt* results in the information contained in the embedded clause being backgrounded.

(43) Cuba and Ürögdi’s (2010) syntactic analysis:

- a. *Péter* (**azt*) *sajnálja*, [_{CP} *hogy havazik*].
Péter (*it.ACC) regrets [_{CP} that it’s.snowing]
‘Péter regrets that it is snowing.’

- b. *Péter (azt_i) mondta, [CP t_i [CP hogy havazik]].*
 Péter (it.ACC) said [CP t_i [CP that it's.snowing]].
 'Péter said that it was snowing.'

Finally, Ürögdi (2008, 2012) and de Cuba and Ürögdi (2010) assume that focused propositional anaphors are generated via a different process syntactically, and not by movement from the embedded clause, which is why they can appear with factives.

Examining similar facts, Molnár (2015) argues that subordinate clauses differ not in reference but in their predication status. The proleptic pronoun *azt* can be found in the modifier position of the matrix verb only if the matrix verb's complement is *predicative*; otherwise, it cannot. According to Molnár's (2015) analysis, a complement is predicative if it has illocutionary force or is in focus. The complements of factive verbs are *non-predicative* in this sense.

4.3 The role of interpretation

Discussing Ürögdi (2008, 2012) and Cuba and Ürögdi's (2010) analysis, Gyarmathy (2012) already pointed out that in certain cases the propositional anaphor *azt* can also appear with so-called factive attitude verbs; however, in these cases the interpretation is non-factive:

- (44) a. *Azt hallottam, hogy Mari megnyerte a lottót.*
 it.ACC heard.1SG that Mari won the lottery.ACC
 'I heard that Mari won.3SG the lottery.'
 b. *Hallottam, hogy Mari megnyerte a lottót.*
 heard.1SG that Mari won.3SG the lottery.ACC
 'I heard that Mari won the lottery.'

"In some cases (I believe primarily with perception verbs), it is not the case that 'azt' is not allowed (or allowed only in a contrastive focus) with factive attitudes, as de Cuba and Ürögdi claim, but it causes a loss of factivity: for example, [(44a)] does not imply that Mari won the lottery, while [(44b)] does. This phenomenon does not question their analysis since it meets the conditions they require. However, the question is whether it could be predicted when *azt* causes a loss of factivity and when it is simply not allowed with factive predicates." - (Gyarmathy 2012, p.56), translation and glosses by me.

Similar examples were seen in (19) and in (20), where the propositional anaphor was in focus or contrastive topic. For this reason, I suggest, in line with Gyarmathy (2012), that the correct generalization is about the interpretation and not about the type of attitude verbs or the type of the complement selected: if the propositional anaphor is in the preverbal modifier position, the interpretation is non-factive, in the sense that the truth of the embedded clause is not necessarily implied and does not project. Thus, the pattern noted by Vendler (1980) in connection with the communicative verb *mond* 'tell', holds more generally as well.

This idea reverses the direction of the argument: Ürögdi (2008, 2012) and Cuba and Ürögdi (2010) assume that factivity is marked in the lexicon and correlates to some extent with the syntactic type of the complement the attitude verb can take. The difference in syntax creates a difference in the informational structure properties observed, namely that in the case of factives the information focus has to be on the verb. However, as was observed by Gyarmathy (2012), they do not explain why some apparently factive verbs can select only CP complements, while other apparent factives have a choice between both types of complements. Further, as it was shown above, focus on the embedded clause sometimes leads to the suspension of projection even with verbs that they consider lexically factive; but this is not explained by their theory. In contrast, I assume that there is no invisible syntactic difference in the type of complements factives and non-factives can take. Instead, differences in information structure and surface morphology lead to the observed differences in veridicality and projection. I argue that information projects when (a) it is not in the information focus of the sentence, (b) its truth is probable and (c) it is independent of the reported attitude. These conditions hold in the case of so-called factive verbs, but not only with these (see, e.g. (20)), and normally do not hold for verbs traditionally considered non-factive, or if the complement of a so-called factive verb is foregrounded via focusing the propositional anaphor.

5 The origins of ‘factivity’

In the previous sections we saw that a number of factors influence whether we get a speaker-related veridical or projective inference with (what are traditionally known as) factive verbs as well as other attitude verbs traditionally not considered factive (e.g. *sejt* ‘suspect’). These factors are (at least):

- (45) a. the type of the proleptic pro-form: *úgy* vs *azt*
- b. preverbal vs. postverbal position of the proleptic pro-form
- c. the presence of focus on the proleptic pro-form
- d. the presence of verbal modifiers such as *jól* ‘well’, *rosszul* ‘incorrectly’
- e. the presence of the perfective particle
- f. the probability of the veridical inference in context

In the following, I show that what looks like a confusing muddle of facts can in fact be explained in a fairly systematic manner. The approach rests on a particular view of presuppositions that regards these as inferences generated by the hearer in a probabilistic fashion.

5.1 Background on presuppositions

When we assert a sentence, our assertion is made in relation to some context: the information state we assume our interlocutors have. The goal of our assertion is to update the

existing information state with new information. Following Stalnaker's (1974) definition, an assertion can be seen as a proposal to update the context.

At the same time, a sentence conveys a lot of information. When we assert something, we are not asserting the complete informational content of the sentence; rather, we are asserting only the information that the presumed information state does not already contain. The information already present in the information state is said to be presupposed. Since the 1970s, it has been observed that certain types of information are regularly presupposed; for example, the complements of so-called factive verbs such as *know* (cf. Kiparsky and Kiparsky 1970). According to the standard theories of presuppositions (Karttunen 1974, Heim 1983, van der Sandt 1992), these presuppositional expressions impose a requirement on the context, namely that the context already contains the presupposed information. The meaning component that imposes this requirement came to be viewed as an independent type of meaning, namely presupposition, as opposed to the entailed content.

In this paper, I approach the problem of presuppositions differently. I start with the premise that there is no categorical distinction between two types of meanings — presupposition and entailment. Therefore, the lexicon does not specify, for example, that the verb *tud* 'know' presupposes its complement. Imagine it did: then we would have to assume that there are factive and non-factive versions of the same verb *tud* 'know', and that the distribution of the two lexical forms depends partly on the pronoun referring to the embedded clause and partly on the position of this pronoun. Further, we would also need to explain that in the presence of evaluative modifiers, the veridicality of verbs can change, while projection disappears. This paper takes a different approach.

When we make an assertion, we do so in relation to an information state: what is new is asserted, and what is old is presupposed. Therefore, what is asserted or presupposed depends in part on the information state, i.e., on what we already know and what is new. In contexts with radically different information bases, the same sentence can assert/presuppose completely different things. This is illustrated by the following example, taken from Stalnaker (1974):

(46) *My cousin is not a boy any more.*

In a context where it is known that the cousin is male but his age is unknown, the sentence asserts that the cousin is now an adult. In a context where the possibility of sex-change is salient, the sentence can assert that the cousin is now female while presupposing that the cousin is still a child. Examples like this show that in theory any piece of information can function as either a presupposition or an assertion. However, there are tendencies and defaults, as well as pragmatic and grammatical constraints, on what type of content is typically presupposed in discourse.

Conversations take place against a background information state (or common ground), Stalnaker (1974), but the interlocutors are unsure what is in the information state of the other participant. Understanding a statement therefore involves the listener trying to infer what the speaker considers part of the common ground. What is 'presupposed' is a hypothesis formed by the hearer about what is in the information state of the speaker

(see partly similar views in: Wilson and Sperber (1979), Qing et al. (2016), Simons et al. (2010), Degen and Tonhauser (2022), Abrusán (2022), Roberts and Simons (2023)).

According to this view, projection is a hypothesis about what part of the information conveyed in the sentence constitutes the background (or common ground) and what part of the information is asserted/negated/questioned, etc. What guides this hypothesis formation? I propose that it is guided by the existing knowledge about the context, but also by our biases about what is likely true in the context, and also by the information structure of the sentence in a broad sense. This view, though a background to the thinking presented in this paper, is also in part motivated by the complex set of facts about Hungarian factives.

What I propose in this paper is that variation in ‘factivity’ is pragmatic in the following sense: the verb *tud* ‘know’ lexically entails subject-related veridicality. This is strengthened to subject+speaker-related veridicality as a default expectation, unless there is morphosyntactic or contextual evidence to the contrary. Similarly, projection of the speaker-related veridical inference is also a defeasible default expectation that is sensitive to the global and morphosyntactic context. Thus, the derivation of speaker-related veridicality and projection are the result of a *multi-factor process* that takes into account lexical meaning, syntax, morphology as well as pragmatic factors such as the likelihood of the veridical (projective) inference and last but not least information structure. The observed veridicality or projection is the result of the interplay of these factors.²¹

This multi-factor approach is in contrast with *silver-bullet approaches* to deriving a factive presupposition, such as QUD-based theories (cf. Simons et al. 2010, 2017), the attention-based theory of Abrusán (2011) or the probabilistic approach outlined in Schlenker (2021). Although these might be on the right track by capturing one or two aspects of the relevant processes needed, they cannot explain the whole picture.

What we need is a theory that can incorporate constraints and pressures of various factors at the same time. In this paper I do not offer a concrete theory of how to derive projection and speaker-related veridicality, but will only sketch a general outline of how such a theory could look like and what factors it should take into account. An example of a concrete theory might be based on the RSA, a probabilistic pragmatics approach that uses structured probabilistic models to develop a general theory of communication (cf. Goodman and Frank (2016), Degen (2023) for an overview, and Degen and Tonhauser (2022) for a similar argument). Preliminary work on presupposition generation was done in this framework by Qing et al. (2016); Warstadt (2022); Scontras and Tonhauser (2024) that looked at predicting the presuppositions of change of state verbs or factives. A related approach is that of Roberts and Simons (2024), who propose that projective inferences of factive verbs can be predicted from the necessary ontological preconditions associated with the event-type they describe. But they also stress that this generalization should be seen as only one factor to be taken into account by a global, more general probabilistic

²¹ A more elaborate formulation of the lexical semantics of so-called factive verbs in Hungarian, presented in Abrusán (2024), proposes that they are underspecified as to the identity of the individual whose evidence is taken to ground the assertion. The value of this variable is set, by default expectation, to be the speaker+hearer+attitude holder, but in the context of certain morpho-syntactic factors (e.g. the presence of *úgy*), the default can be overwritten and specified to be only the attitude holder.

approach to projection.²² Yet, RSA theories also have limitations and it remains to be seen if they can be scaled up to deal with the complex set of facts observed in this paper while maintaining a principled way of setting priors, costs and other factors in a way that is both explanatory and makes correct predictions.²³ It is also possible that in the end a different type of probabilistic approach to linguistic communication (cf. e.g. Erk 2022 for an overview) might be better suited to handle the data presented in this paper.

5.2 Roads to ‘factivity’ in Hungarian

This paper argues that ‘factivity’ is not one property, but a co-occurrence of a set of properties: subject and speaker-related evidential veridicality and projection. Projective inferences are not dependent on the speaker-veridicality of the unembedded positive sentence, though the two typically do co-occur, because they are influenced by an overlapping set of factors. This co-occurrence is what looks like the property of ‘factivity’.

Many attitude verbs have, as part of their semantic meaning in the lexicon, a subject-related evidential veridical entailment. But in unmarked cases, when the veridical inference is based on public evidence, there is a (defeasible) default expectation of a speaker-related veridical inference as well: this is because by default, we assume that given the same evidence speakers, hearers and attitude holders arrive at the same conclusion. This default not being met is the marked case, typically signalled by preverbal modifiers e.g. *úgy* (cf. Section 5.3). The inference that the complement projects from embedded contexts in turn is largely a hypothesis about what the informational state of our interlocutors may contain. What is referred to as a linguistic presupposition is a common type of hypothesis (cf. Section 5.4).

How do conversation participants infer what is in the joint information state (IS) / common ground? They can partly form hypotheses about what IS may contain based on the history of the conversation, partly via their biases and default expectations and partly from the information content of the asserted sentence.

An idea that dates back at least to the Prague School is that assertion, negation, as well as various logical operations target the theme/foregrounded part/main point/at-issue content of the sentence (its aboutness structure), the rest of the information will be backgrounded, as long as this is plausible (cf. also Abrisán 2013). In this paper I explore this idea in the context of Hungarian attitude verbs.²⁴

²²Note that the facts relating to the variation in veridicality presented in this paper pose a challenge to this theory: if ‘factive’ inferences are necessary ontological preconditions for an event-type, then how is it possible that sometimes these ontological preconditions do not hold with the same event-types? Concretely, if the veridical inference of *tud* ‘know’ is an ontological precondition, then why is it absent in some cases?

²³For Qing et al. (2016) for example, the predicted presupposition of change of state verbs ultimately depends on non-trivial assumptions about the nature of the QUD that is assumed.

²⁴A detailed comparison with the ideas of the Prague School concerning allegations and presuppositions, though relevant, is unfortunately beyond the scope of this paper. A reviewer asks if an alternative theory, according to which focalized presuppositions are locally accommodated in entailment-cancelling environments would be viable, as opposed to the idea defended in this paper according to which there are no lexically triggered presuppositions but non-focal propositions tend to be presupposed (cf. Partee 1996).

Crucially, the information content of a sentence is structured: it signals what the sentence is primarily about and what are less important elements of its meaning. Information structuring can arise via topic-focus articulation, prosody or morpho-syntactic means. The idea of such an information structuring has a long history in linguistics and many versions of it have been proposed under various names: theme-rheme or topic-focus articulation (Prague School, cf. e.g. Hajičová et al. 1998), foregrounded-backgrounded information, information focus, main point vs. not main point (cf. e.g. Abrusán 2011), at-issue vs. not at-issue meaning (cf. Koev 2018), the subject matter (Yablo 2014, Fine 2017) etc. Though related, these are all subtly different notions: for example while the notion of *at-issueness*, following Simons et al. (2010), is often understood to be the information that answers an implicit QUD, the *main point* as defined in Abrusán (2011) is determined by the event structure of the complex event described by the sentence. In this paper I use the notions *main point* or *information focus* in Hungarian in a syntactically determined way to refer to the information that is in the preverbal field of the Hungarian sentence (or strictly related to it). This notion, unlike typical (though not all, see Koev 2018) understandings of at-issueness, does not necessarily relate to an implicit QUD. Elements in the preverbal field might be answering implicit questions, but this is by no means always the case.

Note that the question of whether or not there are lexically factive predicates is separate from another issue, namely whether projection is a categorical or a gradient property, as argued by Tonhauser et al. (2018).²⁵ As far as I can see, the Hungarian data presented above have no bearing on this question: they are compatible with a view on which a probabilistic mechanism outputs a projective inference with a certain degree of strength, but also with a view on which projection is understood as categorical for any given example/speech situation, but hearers are unsure whether a projective or a non-projective reading should be derived (cf. Grove and White 2024). The projective inference derived by a pragmatic process for any particular example could be categorical, as long as the possibility of variation in projection for the same predicates is allowed.

5.3 Which inferences are speaker-veridical?

The meaning of most verbs traditionally considered factive lexically only implies a subject-related veridical inference in Hungarian: Namely, that the subject of the attitude believes that the complement is true and that this belief is based on evidence available to the subject and perhaps other factors.

But by default, a speaker-related veridical inference is also drawn from the sentence in these cases, at least if further grammatical and pragmatic factors are present: (a) backgrounding the complement clause, which is signalled by the propositional anaphor not being in the preverbal modifier position (though it might be focused), (b) the particles *el-*, *meg-* might impose a specificity requirement on their object, which can be understood

However, a theory based on local accommodation would not be able to derive the facts with *úgy* ‘so’ and *rosszul* ‘badly’ nor other fluctuations in veridicality. Local accommodation is also a rather ill-understood process, in general.

²⁵Thanks to an anonymous reviewer who suggested I clarify this issue.

as discourse oldness/backgrounding. Conversely, introducing the complement clause by the hearsay evidential maker *úgy* ‘so’ or modifying it by the evaluative adverbial *rosszul* ‘incorrectly’ signals that the speaker does not take the complement clause to be true.

The above also interact with pragmatic factors: the reliability of the subject, which can imply that their justified beliefs might actually be true, and the contextual probability of the truth of the complement in question.

Examples Let’s illustrate the above informal reasoning with a few examples:

- (47) *Mari* [_{PredP} [_{Pred'} *tudja*_j [_{VP} *t_j* (*azt_i*)]]], [_{CP} *hogy esik*]_i
Mari [_{PredP} [_{Pred'} *knows_j* [_{VP} *t_j* (*it.ACC_i*)]]], [_{CP} *that rains*]_i
 ‘Mari knows that it is raining.’

Let us informally assume here²⁶ that the lexical meaning of the verb *tud* is the following:

- (48) *a tudja that p* is true if
 a. *a* believes that *p* and
 b. *a* has enough evidence for asserting *p*

The lexical meaning thus entails that the subject has sufficient evidence for *p*. In (47), the verb is in the head position of PredP, and the propositional anaphor *azt*, which can also remain silent, is in a postverbal position. This suggests that the main point of the sentence is that Mari has a certain propositional attitude, and not the content of her beliefs. The verb *tud* ‘know’ syntactically cannot appear with an unfocused preverbal propositional anaphor, unlike the verb *gondol* ‘thinks’, e.g., for which this is the default syntactic configuration (cf. Ürögdi 2012 for ample discussion.)

- (49) a. **Mari* [_{PredP} [_{Pred'} *azt tudja*_j [_{VP} *t_j* (*t_i*)]]] [_{CP} *hogy esik*]_i
Mari [_{PredP} [_{Pred'} *it.ACC knows_j* [_{VP} *t_j* (*t_i*)]]] [_{CP} *that rains*]_i
 ‘Mari knows that it is raining.’
 b. *Mari* [_{PredP} [_{Pred'} *azt gondolja*_j [_{VP} *t_j* (*t_i*)]]] [_{CP} *hogy esik*]_i
Mari [_{PredP} [_{Pred'} *it.ACC thinks_j* [_{VP} *t_j* (*t_i*)]]] [_{CP} *that rains*]_i
 ‘Mari thinks that it is raining.’

The unavailability of drawing attention syntactically to the complement clause suggests that the complement is normally backgrounded and not new material, and hence that *the speaker* does not treat it as controversial information, presumably because they have evidence for it themselves. As a result, the inference arises that the speaker assumes that the content of the complement is true. As this is the most frequent situation, the verb is felt to be speaker-veridical, in the unmarked case. Note also that the speaker-related veridical inference is not easy to cancel (for example by simply following up (47)

²⁶See Abrusán (2024, 2025) for a detailed formal analysis of the meaning of (non-factive) *know* in Hungarian. Explicating the formal analysis here would take us too far from the main aim of this paper, which is understanding the factors for variations in factivity.

with ‘But it is not raining’), so this is not a run-of-the-mill conversational implicature. I propose that the veridical reading of *tud* is a *default reading* in the sense of Roberts and Simons (2024): a probabilistically expected default reading that is, however, defeasible given enough countervailing syntactic, morphological and contextual cues.²⁷

Why not assume that *tud* semantically entails the truth of its complement? The reason is the existence of examples with the evidential marker *úgy* and the modifier *rosszul*. If these were to combine with a veridical *tud*, composition would have to involve the subtraction of information present in the lexical items.²⁸

- (50) a. *Mari* [_{PredP} [_{Pred'} *úgy tudja_j* [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 Mari [_{PredP} [_{Pred'} *so knows_j* [_{VP} *t_j(t_i)*]]] [_{CP} *that rains*]_i
 ‘*Mari* (lit.) so-knows that it is raining.’
 b. *Mari* [_{PredP} *rosszul* [_{Pred'} *tudja_j* [_{VP} *t_j(azt_i)*]]] [_{CP} *hogy esik*]_i
 Mari [_{PredP} *wrongly* [_{Pred'} *knows_j* [_{VP} *t_j(it_i)*]]] [_{CP} *that rains*]_i
 ‘*Mari* (lit.) incorrectly knows that it is raining.’

In (50a) the evidential marker shows that the speaker has no independent evidence for the complement: the speaker’s source of information is *Mari*. Thus, the truth of the complement is not entailed by the positive sentence. In (50b) the result of the adverbial modification is that the truth of the complement is not implied.

One might be tempted to think that the position of the propositional anaphor (or the (invisible) difference in the type of the associated complement/LF) encodes veridicality in a hard-wired manner. This would mean that the post-verbal propositional anaphor is somehow associated with veridical (and projective?) complements. (Analyses in this vein were proposed for factivity alternations in Turkish and Azeri (cf. Özyildiz (2017b,a); Bondarenko (2023), see Section 2.7.). However, this cannot be the case in Hungarian; for example (50b) is compatible with post-verbal *azt*, despite a clearly non-veridical reading. (Cf. also arguments in Ürögdi 2012 for why the type of the complement cannot be mapped in a one-to-one manner to factive/non-factive interpretations: the most important being that non-factives can often take both types of complements.)

I propose, thus, that the veridical reading of *tud* is a default reading, the one we get in the absence of countervailing cues. With syntactic cues such as *úgy* and *rosszul*, a non-veridical reading arises. As discussed in Roberts and Simons (2024), the often-assumed dichotomy between conventional and pragmatic meaning is an oversimplification:

²⁷The view presented here also shows some similarity with that of Hazlett (2010), who derives the veridical inference of the English verb *know* as a pragmatic inference that can be drawn in some (but not all) contexts.

²⁸This situation is not unheard of: this is what happens in the case of modification with privative adjectives, e.g. *fake gun*. The classical approach to this problem, due to Partee (1995), assumes that the privative adjective *fake* coerces the noun *gun* into meaning ‘gun-like entities’. I think that a similar approach to Hungarian ‘factives’ would not be attractive, mainly because the non-factivity does not only appear during the composition with *úgy*, *rosszul* ‘so, badly’, but also via alternating the syntactic position and the discourse function of the propositional anaphor *azt* ‘it.ACC’, especially with verbs such as *lát* ‘see’, *sejt* ‘suspect’, etc.

pragmatic meaning can arise in many ways and can have varying properties, and it can be more or less easily defeasible depending on the particular circumstances. Defaults might behave as conventional meaning, for all intents and purposes.

In the case of certain perceptual and epistemic verbs (e.g., *hall* ‘hear’, *sejt* ‘suspect’, cf. Section 3.1), the proleptic pronoun *azt* ‘it.ACC’ can appear before or after the verb. In the first case, the assertion applies to both the matrix verb and the embedded clause (51a). In such cases, the veridical inference depends on how much the listener considers Mari (and the informants) a reliable source. However, if *azt* is omitted or is postverbal as in (51b), the main assertion applies only to the matrix verb. Since the main assertion does not concern the subordinate clause, the listener can infer that the embedded clause is independent of the assertion, for example, because the speaker has independent evidence for it. In this case, if the context also makes the truth of the subordinate clause likely, a veridical inference arises.

- (51) a. *Mari* [_{PredP} [_{Pred'} *azt*_i *hallotta/sejt*_i [_{VP} *t*_j (*t*_i)]]] [_{CP} *hogy esik*]_i
 Mari [_{PredP} [_{Pred'} it.ACC_i heard/suspects_j [_{VP} *t*_j (*t*_i)]]] [_{CP} that rains]_i
 ‘Mari heard/suspects that it is raining.’
 b. *Mari* [_{PredP} [_{Pred'} *hallotta/sejt*_i [_{VP} *t*_j (*azt*_i)]]] [_{CP} *hogy esik*]_i
 Mari [_{PredP} [_{Pred'} heard/suspects_j [_{VP} *t*_j (it.ACC_i)]]] [_{CP} that rains]_i
 ‘Mari heard/suspects that it is raining.’

Thus, in (51a) a speaker-veridical inference, if there is one, is based entirely on the context (on the reliability of the subject, etc.) The example in (51b), however, is more strongly speaker-veridical, as syntactic structure suggests that the complement should be backgrounded. (Speaker-veridicality also depends on the position of the propositional anaphor in the case of many emotives, as was shown in Section 4.2, cf. example (29).)

The following examples illustrate the impact of perfective verbal particles on speaker-veridicality. In the case of communicative verbs such as *mond* ‘say’ or *ír* ‘write’, when the proleptic pronoun is preverbal and there is no perfective particle, we get a non-veridical reading, cf. (52a). However, the version with the verbal particle *megmond* is speaker-veridical cf. (52c). The reading of the example in (52b) can be speaker-veridical or not, depending on how likely the context makes the truth of the embedded clause (regardless of what Mari said).

- (52) a. *Mari* [_{PredP} [_{Pred'} *azt*_i *mondta*_j [_{VP} *t*_j *Jancsinak t*_i]]], *hogy...*
 Mari [_{PredP} [_{Pred'} it.ACC_i said_j [_{VP} *t*_j Jancsi.DAT *t*_i]]], that...
 ‘Mari told Jancsi that...’
 b. *Mari* [_{PredP} [_{Pred'} *mondta*_j [_{VP} *t*_j *Jancsinak (azt*_i *)*]]], *hogy...*
 Mari [_{PredP} [_{Pred'} said_j [_{VP} *t*_j Jancsi.DAT (it.ACC_i)]]], that...
 ‘Mari told Jancsi that...’
 c. *Mari* [_{PredP} [_{Pred'} *meg mondta*_j [_{VP} *t*_j *Jancsinak (azt*_i *)*]]], *hogy...*
 Mari [_{PredP} [_{Pred'} PRT said_j [_{VP} *t*_j Jancsi.DAT (it.ACC_i)]]], that...
 ‘Mari told Jancsi that...’

Note also that the verbal particle *meg* introduces a complex event structure and draws attention on the moment of state change. Since the object of the verb with the verbal particle *meg* must be [+specific], the event described by the embedded clause must already exist before the change of state. Moreover, the postverbal position of the propositional anaphor signals that the complement is backgrounded. From these two factors hearers can deduce that the truth of the complement is uncontroversial, hence likely true in the context.

Finally, note that focusing the propositional anaphor does not itself interfere with the expectation of default speaker-related veridicality in Hungarian. If does, however, interact with projection, as shown below.

5.4 What information projects?

Projection, I assume, is the consequence of the context-sensitivity of logical operators such as negation or conditionals. As a result of this context-sensitivity, operators target only part of the information conveyed by the sentence, and not all of it. How do we know which part of the total information conveyed is (not) targeted by the logical operators (negation, conditionals, etc.)? In fact, we do not know, but we infer this based on the history of the conversation, contextual factors, the content of the sentence and various syntactic, morphological and prosodic factors (and more). Thus, projection is largely a hypothesis about what the informational state may contain. What is referred to as presupposition is a common type of hypothesis.

For example, negating a complex proposition is compatible with negating only part of it, i.e. it is not necessary for every consequence of the complex proposition to be false; it is sufficient if one of the consequences is false. For example, in the case of $\neg(A+B)$, it is enough if either A is false or B is false. Natural language negation almost never applies to all consequences of a sentence; only to some of them (cf. Kratzer 1989, and subsequent discussion on the focus-(in)sensitivity of negation). Which ones? Most often, negation targets the main point of the sentence, what the sentence is about. Information that is backgrounded in some way (e.g. by syntactic or prosodic means, or by being contextually likely to be true) is not affected by negation and is assumed to be part of the common information state. Thus consequences of a sentence that are likely part of the common ground do not need to be affected by the negation. As a result, the semantic scope of negation is determined based on the aboutness structure of the sentence and also based on considerations of plausibility: it will include information that is not part of the main point of the sentence, and that is likely true.²⁹ This view captures the context-sensitive nature of negation in natural language: it rules out states that are incompatible with what we know to be the case.³⁰

²⁹As is well known, projection does not only happen from the scope of negation but also from the scope of other entailment-cancelling operators. I am currently developing an approach similar to negation based on the context-sensitivity of conditionals and other operators. Sketching this account here would take us too far.

³⁰This way of thinking of negation, though classical, shares some of the motivations for non-classical

In Hungarian, the main point of the sentence is related to the matrix verbs and the pre-verbal field in PredP (see also discussion in É. Kiss 2008a). This is the centre of attention. Negation (and other operators) target the main point of the sentence, e.g. negation negates what is in the head and Specifier of PredP. Intuitively, if the centre of attention only contains the main verb, negation targets that. If a propositional anaphor is also in PredP, negation will also target the complement (the effect is also modulated by focus, as will be shown below). If a verbal modifier (*jól* ‘well’) is in PredP, negation will target that as well. If a perfective particle whose function is to introduce complex events is in PredP, negation targets the last component of the event, but the rest of the information projects.

The centre of attention in Hungarian [i.e the specifier and head of PredP] reflects to some extent canonical Information Structure (i.e. focus-topic organisation), but also the conceptual structural organisation of the content (*aboutness*, cf. Yablo 2014). As a result, what projects depends both on the information structure (QUD, cf. Simons et al. 2010) and on the syntactic and conceptual structure (cf. Abrusán 2011).

Further, information can only project if it is *independent* of the main point, at least in the sense of being able to be true while the main point is false. Slightly more precisely, I assume that the property of independence is understood as being grounded by different events/states (cf. Abrusán 2011, Yablo 2014). Finally, projection also interacts with the reliability of the subject and the probability of the truth of the complement (cf. Degen and Tonhauser 2021).

Projection, thus, is a probabilistic process. What projects is the information that we think is likely true and is independent of the main point and we don’t pay attention to. Projective inferences are not dependent on the veridicality of the positive sentence, though the two typically do co-occur, because they are influenced by an overlapping set of factors. This frequent co-occurrence of an (objective) veridical inference and projection is what has been called ‘factivity’.

Examples Consider a simple example of negated knowledge ascription. NB: I use syntactically wide-scope negation to avoid ambiguity between sentence and constituent negation.

- (53) *Nem igaz, hogy Mari* [_{PredP} [_{Pred'} *tudja_j* [_{VP} *t_j* (*azt_i*)]]], [_{CP} *hogy esik*]_i
 not true that Mari [_{PredP} [_{Pred'} knows_j [_{VP} *t_j* (it.ACC_i)]]], [_{CP} that rains]_i
 ‘It is not true that Mari knows that it is raining.’

In this case, the main point is in the PredP of the matrix sentence, the propositional attitude of Mari. Negation targets the main point, the propositional attitude of Mari. The information that is not part of the main point (that it is raining) and is logically independent of it is not affected by negation. (Note that various other entailments (e.g. somebody thinks it is raining, etc.) are not independent of the main point, cf. Abrusán (2011).) The IS is then updated with the negation of the main point. As the secondary point is grounded by independent facts, and is not targeted by negation, its non-negated

views of negation understood as incompatibility, similarly to the modal negation view in Berto (2015).

content is added to the IS unless it is highly unlikely in the context (if it is already part of the IS).

Observe now the following example with the evidential marker *úgy*:

- (54) *Nem igaz, hogy Mari* [_{PredP} [_{Pred'} *úgy tudja_j* [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 not true that Mari [_{PredP} [_{Pred'} so knows_j [_{VP} *t_j(t_i)*]]] [_{CP} that rains]_i
 ‘It is not true that Mari (lit.) so-knows that it is raining.’

The evidential marker shows that the speaker’s source of information is Mari and there are no independent facts that ground the truth of the complement other than Mari’s attitude. Thus, the truth of the complement is not entailed by the positive sentence and the preverbal position of the pro-form *úgy* also marks that the complement is part of the main point. Negation, as before, targets the main point, however, in this case the information in the embedded clause is syntactically part of the centre of attention and is semantically not independent from the main point and therefore it is also targeted by negation. There is no projective inference in this case. The situation is thus similar to what we find with a verb such as *gondol* ‘think’:

- (55) *Nem igaz, hogy Mari* [_{PredP} [_{Pred'} *azt_i gondolja_j* [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 not true that Mari [_{PredP} [_{Pred'} it.ACC thinks_j [_{VP} *t_j(t_i)*]]] [_{CP} that rains]_i
 ‘It is not true that Mari thinks that it is raining.’

In the above example, the preverbal position of the propositional anaphor *azt* indicates that the attention is directed not only to the matrix verb but also to the embedded clause. Therefore, the embedded clause is also part of the main assertion. The truth of the embedded complement will not project in the scope of negation. In the case of certain perceptual and epistemic verbs (e.g., *hall* ‘hear’, *sejt* ‘suspect’), the pronoun can appear before or after the verb. In the first case, the centre of attention includes both the matrix verb and the embedded clause (56a). In such cases, there is no projection, and the veridical inference depends on how much the listener considers Mari (and the informants) a reliable source. However, if *azt* is omitted or is postverbal as in (56b), the main assertion applies only to the matrix verb. Since the main assertion does not concern the subordinate clause, the listener can infer that the embedded clause is independent of the assertion, for example, because the speaker has independent evidence. In this case, if the context also makes the truth of the subordinate clause likely, we observe projection, i.e. (56b) suggests that it is raining.

- (56) a. *Nem igaz, hogy Mari* [_{PredP} [_{Pred'} *azt_i hallotta/sejti_j* [_{VP} *t_j(t_i)*]]] [_{CP} *hogy*
 not true that Mari [_{PredP} [_{Pred'} it.ACC heard/suspects_j [_{VP} *t_j(t_i)*]]] [_{CP} that
esik]_i
 rains]_i
 ‘It is not true that Mari heard/suspects that it is raining.’
 b. *Nem igaz, hogy Mari* [_{PredP} [_{Pred'} *hallotta/sejti_j* [_{VP} *t_j(t/azt_i)*]]] [_{CP} *hogy*
 not true that Mari [_{PredP} [_{Pred'} heard/suspects_j [_{VP} *t_j(t/it_i)*]]] [_{CP} that

esik]_i
 rains]_i

‘It is not true that Mari heard/suspects that it is raining.’

If the pronoun referring to the embedded clause is in a focus position, there are several possible interpretations depending on the context. Let’s first consider the following case.

(57) *Context 1*: The information state includes that it is raining and that it is windy. The relevant question is: ‘What does Mari know from all this?’

- a. *Mari* [_{PredP} *AZT*_i [_{Pred'} *tudja*_j [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 Mari [_{PredP} it.ACC [_{Pred'} knows_j [_{VP} *t_j(t_i)*]]] [_{CP} that rains]_i
 ‘What Mari knows is that it is raining.’
- b. *Nem igaz, hogy Mari* [_{PredP} *AZT*_i [_{Pred'} *tudja*_j [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 not true that Mari [_{PredP} it.ACC [_{Pred'} knows_j [_{VP} *t_j(t_i)*]]] [_{CP} that rains]_i
 ‘It is not true that what Mari knows is that it is raining.’

Focus on the pronoun *azt* foregrounds the information in the embedded clause. In the positive sentence in (57a), this content is therefore asserted, but it has no effect on the context since in the example above Context 1 already includes this information. In the negative sentence in (57b), the content of the embedded clause is part of the denied content. However, this content is complex, and it is not necessary for every part to be false for the denial to be true: this is because $\neg(A \wedge B)$ does not imply that both $\neg A$ and $\neg B$; it is enough if one of the two is false. If the context already implies that it is raining, this content is compatible with the denial of the complex sentence. In a different context, however, we get a different result:

(58) *Context 2*: The information state does not contain any other information, only the question ‘What does Mari know?’

- a. *Mari* [_{PredP} *AZT*_i [_{Pred'} *tudja*_j [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 Mari [_{PredP} it.ACC [_{Pred'} knows_j [_{VP} *t_j(t_i)*]]] [_{CP} that rains]_i
 ‘What Mari knows is that it is raining.’
- b. *Nem igaz, hogy Mari* [_{PredP} *AZT*_i [_{Pred'} *tudja*_j [_{VP} *t_j(t_i)*]]] [_{CP} *hogy esik*]_i
 not true that Mari [_{PredP} it.ACC [_{Pred'} knows_j [_{VP} *t_j(t_i)*]]] [_{CP} that rains]_i
 ‘It is not true that what Mari knows is that it is raining.’

If nothing in the context implies the truth of the embedded clause, as in the example above, in the positive case in (58a), both the content of the main clause (i.e., the propositional attitude) and the content of the embedded clause (i.e., that it is raining) are asserted. In the negative case (58b), all the information conveyed by the sentence is negated because the pronoun referring to the embedded clause is in PredP. However, no inference can be drawn about the truth of the embedded clause (as it can be either true or false). Therefore, in this case, we do not see a projective inference. If, however, it is already in the speaker’s informational state that it is not raining, then the negative sentence is most easily interpreted as denying both the embedded clause and the matrix attitude.

Observe now an example of *tud* ‘know’ with preverbal modifiers:

- (59) *Nem igaz, hogy Mari* [_{PredP} *JÓL/ROSSZUL* [_{Pred'} *tudja_j* [_{VP} *t_j(t_i)*]]] [_{CP} *hogy*
 not true that Mari [_{PredP} well/badly [_{Pred'} knows_j [_{VP} t_j(t_i)]]] [_{CP} that
esik]_i
 rains]_i
 ‘It is not true that Mari is correct/incorrect that it is raining.’

The modifiers *jól* ‘well’ and *rosszul* ‘badly’ indicate that the truth of the embedded clause is questionable in the context. Since these modifiers are in focus position, attention is directed to the subordinate clause, making the subordinate clause part of the assertion. In this case, we do not see projection in the scope of entailment-cancelling operators. However, a non-embedded attitude ascription is veridical with the verb modifier *jól* ‘well’ and anti-veridical with the verb modifier *rosszul* ‘badly’, because, in this case, the sentence answers the question present in the context.

The following examples illustrate the impact of perfective verbal particles on projection. In the case of communicative verbs such as *mond* ‘say’ or *ír* ‘write’, when the proleptic pronoun *azt* is preverbal and there is no perfective particle, we get a non-projective reading in (60a). However, the version with the verbal particle *megmond*, *megír* in (60c) is both veridical and projective. The reading of the example in (60b) can be projective or not, depending on how likely the context makes the truth of the embedded clause (regardless of what Mari said).

- (60) *Nem igaz, hogy...*
 not true that
 ‘It is not true that...’
- a. *Mari* [_{PredP} [_{Pred'} *azt_i* *mondta_j* [_{VP} *t_j Jancsinak t_i*]], *hogy...*
 Mari [_{PredP} [_{Pred'} it.ACC said_j [_{VP} *t_j Jancsi.DAT t_i*]], that...
 ‘Mari told Jancsi that...’
- b. *Mari* [_{PredP} [_{Pred'} *mondta_j* [_{VP} *t_j Jancsinak (azt_i)*]]], *hogy...*
 Mari [_{PredP} [_{Pred'} said_j [_{VP} *t_j Jancsi.DAT (it.ACC_i)*]]], that...
 ‘Mari told Jancsi that...’
- c. *Mari* [_{PredP} [_{Pred'} *meg mondta_j* [_{VP} *t_j Jancsinak (azt_i)*]]], *hogy...*
 Mari [_{PredP} [_{Pred'} PRT said_j [_{VP} *t_j Jancsi.DAT (it.ACC_i)*]]], that...
 ‘Mari told Jancsi that...’

Note also that the verbal particle *meg* introduces a complex event structure and draws attention to the moment of state change. Negation relates to this change of state. Since the object of the *meg* verbal particle must be [+specific], the event described by the embedded clause must already exist before the change of state, requiring a specific interpretation. It follows that the truth of the embedded clause must already follow from the informational state. The projective reading, in this case, is a consequence of the anaphoric requirement of the *meg* verbal particle.

6 Perspective shift

In this paper I made a distinction between subject-related and speaker-related veridicality. The first one, speaker-related veridicality, involves a certain shift towards the perspective of the subject talked about, *úgy* signalling that the attitude holder’s standards for evidence imply that the complement is true.

A silent move to the attitude holder’s perspective is also possible without *úgy*, but this needs a rich supporting context and special intonation that suggests that we are moving to a special mode of talking (perhaps what has been called *protagonist projection*, cf. e.g. Holton 1997, Abrusán 2022). Such shifts are facilitated by a pre-verbal position of the pronoun *azt*, though this is not always required, as will be shown below. Observe first the example (61), in which the pronoun is focused. In this example the subject’s attitude is reported as knowledge, but the speaker demonstrates a certain distance from the subject’s stance.

- (61) *De nem csak azt tudja Kósa, hogy Budapest jobb hely Bécsnél, hanem azt is, hogy a CEU csak mórlikálja magát, magyarul, játssza az eszt, miközben esze ágában sincs Bécsbe költözni, csak fut még egy kört.*
 but not only it knows Kósa that Budapest better place from.Vienna but
azt is, hogy a CEU csak mórlikálja magát, magyarul, játssza az eszt,
 it too that the CEU only pretend.plays itself, in.Hungarian, plays the mind,
miközben esze ágában sincs Bécsbe költözni, csak fut még egy kört.
 while mind’s branch.in isn’t to.Vienna move.INF, only run more one round.
 ‘But not only does Kósa know that Budapest is a better place than Vienna, he also knows that the CEU is just playing pretend games, that is, it is just trying to act smart, while it has no intention of moving to Vienna, just running another round.’
 (<https://www.ceu.edu/article/2018-11-28/kosa-lajos-ugy-tudja-hogy-ceu-budapesten-marad> (accessed on January 7, 2025))

In the above example, Kósa’s attitude to the CEU is reported, but the author of the description absolutely does not imply that the CEU is playing a pretend game or trying to act smart, if anything, the contrary is suggested. Observe another example, in this case the pronoun is in contrastive topic position, shown by the required intonation pattern and the short pause after *azt*.

- (62) *Kovács Ákos saját bevallása szerint sem a háború szakértője, de azt tudja: aligha az oroszok érdeke az Európa ellen folyó aljas játszma.*
 Kovács Ákos own admission according-to neither the war expert.POSS but
azt tudja: aligha az oroszok érdeke az Európa ellen folyó
 it.ACC knows: hardly the Russians interest.POSS the Europe against ongoing
aljas játszma.
 vile game
 ‘Even by his own admission, Ákos Kovács is no expert on war, but he does know that the vile game against Europe is hardly in the Russians’ interest.’
 (<https://444.hu/2022/12/24/kovacs-akos-sajat-bevallasa-szerint-sem-a-haboru-szakertoje->

de-azt-tudja-aligha-az-orosz-erdeke-az-europa-ellen-folyo-aljas-jatszma (accessed on January 7, 2025))

The example above is not speaker-veridical: The context makes it clear that the author of the sentence is clearly not committed to the truth of the complement, namely that the vile game against Europe is hardly in the Russians’ interest. In fact, the sentence suggests that the author does not consider the subject, Ákos Kovács (a well-known Hungarian pop-singer), to be a reliable source for the question of the truth of the complement.

In cases of extreme unreliability of the subject, e.g. in the context of talking about mental illness, or other contexts that license shifting to the perspective of the subject, it is possible to get subjective readings of *tud* without the proleptic pronoun as well, with stress on the verb:

- (63) *Próbáltam beszélni a beteggel, de nem sikerült meggyőzőnöm. Ő*
 tried.1SG talking the patient.WITH, but not manage convince.1SG.DEF (s)he
tudja, hogy ő Jézus, és meg kell váltania a világot.
 knows that (s)he Jesus and PRT has.to redeem the world
 ‘I tried talking with the patient, but I did not manage to convince him/her. (S)he knows that (s)he is Jesus, and that (s)he needs to redeem the world.’³¹

In the above cases, there is a feeling that the verb *know* itself is in the scope of an external perspective shift that reports a protagonist’s discourse or perception, similar to free indirect discourse or protagonist projection (cf. Abrusán 2022). The report is felt to be almost literal in the last case (63), which is understood as reporting almost verbatim the patient’s speech or thoughts “I know I am Jesus and I need to save the world”. This reportative aspect is much less present in the previous examples, which do not report verbatim their subjects saying or thinking “I know that p”: they only report that the subject manifested (in some way) knowledge of p. The discussion in (61) is presumably only based on Kósa saying that the CEU does not really want to move, and not on him saying that he knows this. The distinction between the two cases is subtle though, especially in the light of the discussion on *protagonist projection*, as the latter can be licensed in more indirect ways than free indirect discourse, cf. Abrusán (2021).

7 Conclusion

What is factivity? According to one extreme, it is a lexical property of certain predicates (Kiparsky and Kiparsky 1970), while according to the other extreme, it is a myth (Hazlett 2010).

Based on the empirical properties of so-called ‘factive’ verbs in Hungarian, I argued that what is usually called ‘factivity’ is not a single property but a set of properties: the simultaneous occurrence of a speaker-related evidential veridical inference, a subject-related

³¹Hungarian pronouns are gender neutral.

evidential veridical inference and the projection of the speaker-related veridical inference from the scope of entailment-cancelling operators. These properties can be manifested separately in various morpho-syntactic contexts. Some of these contexts signal a default, speaker+subject-related evidential veridical interpretation, while certain elements, e.g. *úgy*, mark departure from this default interpretation and signal only subject-related evidential veridicality. It follows that ‘factivity’, or even ‘veridicality’, at least in Hungarian, is not lexically encoded as a property of so-called ‘factive’ verbs.

I also argued that the projective inference is a hypothesis formulated by the listener about the speaker’s informational state. I outlined how this hypothesis formation is influenced by syntax, certain morphological factors, information structural considerations, and the probability of conveyed information in context. Finally, I also argued that the projection of veridical inference in an entailment-cancelling environment is a byproduct of the context-dependence of natural language operators.

Acknowledgements

This draft is partly based on a previous paper written in Hungarian by the author (Abrusán 2023); some paragraphs are close translations of paragraphs in that earlier paper. I am grateful to the editors, Edgar Onea and Balázs Surányi as well as three anonymous reviewers for extremely helpful suggestions and criticisms that helped improve the paper greatly. Thanks also to the audiences at the University of Geneva, University of Uppsala, Sinn und Bedeutung 29 in Noto and the Amsterdam Colloquium 2024 as well as Dean McHugh and Yoad Winter for their questions concerning this or related material. Finally, thank you to Péter Siptár for his thorough proofreading of the text.

Funding

The research reported here was supported by the institutional grant ANR-17-EURE-0017.

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